Real Mode
- Segment
- Offset

→ 1MB

Protected Mode
- Selector
- Descriptor

Q. Real mode vs Protected Mode
Q. Global vs Local descriptors

Total descriptors
(both local & global
16,384)

↗ Protected Mode

- ✗ No segment register
- ✓ Operates above the first 1MB allocated real memory
- ✓ Used in Windows operating system.
- ✓ "Selector" selects a descriptor

[Segment Register]
↓
Selector
↓
Select descriptor from descriptor table
↓
local global

↗ Contents of descriptor
- ↳ Segment location
- ↳ Length
- ↳ Access Rights

[Selector is located in the segment register]
→ Selects one of 8192 descriptors from 1/2 descriptor tables

Local descriptors are unique to application

Global descriptor ~ System Descriptor

Local descriptor ~ Application Descriptor

31		0
0000 0000 0000 0000	Address Rights 8bit	Base 8bit B23 B16
Base 16bit B15 B0	Limit 16bit L15	L0

base = 16 bit + 8bit = 24 bit

limit = 16 bit

~~excess~~

Access right = 8 bit

table 2 [B Page 64]

figure 2-6

Core 2 → each descriptor is 8 bytes in length.

Maximum in descriptor table = 64K

Base address denotes starting location.

↳ 24 bits

Bits → Granularity bit

If G=0,

If G=1,

↗ Example 2.1

Base = Start = 10000004

G=0

End = Base + Limit = 10000000H + 001FFH

= 1000 1FFH

hasan

↗ Example 2.2

$G = 1$

Base = same as before

End = Base + Limit (append + FFF)

= 1000000H + 001FFFFF

= 101FFFFFH

Bits = L & AV

L = 1 → 64 bit
L = 0 → 32 bit → compatibility mode

→ 64 bit descriptor

AV ~ is segment available

AV = 0 → unavailable

AV = 1 → segment available

Bits = D

D = 0 → 16 bit instruction

D = 1 → 32 bit instruction

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RPL 00 → highest
01
10
11 → lowest

CS = 0008H

15 ... 3 2 1 0

0000 0000 0000 1 0 00

→ RPL → requested previllage level

→ TI ⇒ 0 = Global, 1 = Local

→ Selects of of the descriptors from 8192 choices

Shayan

DPL

↳ Descriptor Privilage ~~Limit~~ Level

	00	
RPL =	01	☑ RPL approved
DPL =	10	
	11	

Class Work

CS = 002CH

00000000000101|100

→ RPL = 00

→ TI = local

→ Descriptor = 5

~ Descriptor ~

00 00 92 10 00 00 00 FF

~~82006~~

0000	92	10
0000	00FF	

80286

Shayan

92

1	0 0	0	1	0	1	0
P	DPL	S	E	ED/C	R/W	A

13 selector chooses → descriptor

→ if the requested privillage level matches or higher in priority than ...

Microcontroller – 2 set questions
Microprocessor – 2 set questions